

· 指南与共识 ·

中国胰腺癌病理取材及诊断报告规范循证学指南

中国医师协会胰腺病学专业委员会 国家消化系统疾病临床医学研究中心(上海)

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【提要】 胰腺癌标本的病理学分析最主要的目的是通过评估肿瘤的组织学类型、分化程度、肿瘤大小、浸润范围、切缘状态等以评估肿瘤的分期及预后,而完成精准评估的首要前提就是病理取材的细致和全面。本指南就胰腺癌病理取材、切缘定义、结构化报告内容和诊断要点等形成共识,旨在促进我国胰腺癌取材及诊断报告的规范化、精准化,提高病理医师精准诊断的能力,为临床提供切实有效的评估预后的病理学参数,从而为患者后续治疗方案的制订提供有力的依据。同时标准化的肿瘤数据收集也将为流行病学专家提供信息,有助于国内及国际规范的制定及研究。

【关键词】 胰腺癌; 病理取材; 切缘; 病理报告; 结构化; 规范化

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Evidence-based guidelines for standardized pathologic sampling and diagnosis reporting of pancreatic cancer in China

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【Summary】 The main purpose of pathological analysis of pancreatic cancer(PC) specimens is to assess the tumor stage and predict the prognosis by evaluating the histological type, degree of differentiation, tumor size, depth of invasion, surgical margin status and the like. Careful, standardized and comprehensive pathological specimen sampling is prerequisite for accurate assessment. This guideline compiles a consensus on pathological sampling, definition of surgical margins, content and diagnostic points of structured report for PC after discussion, aiming to improve the standardization and precision of PC specimen sampling and diagnostic reports in our country, and enhance the precision diagnostic ability of pathologists. These provide clinicians with practical and effective pathologic parameters for prognosis assessment, thereby providing strong support for the formulation of treatment strategies for patients after surgery. Standardized oncology data collection also provides information for epidemiologists, which contributes to national and international standardization and research.

【Key words】 Pancreatic cancer; Pathological specimen sampling; Surgical margin; Pathologic report; Structuration; Normalization

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成人胰腺肿瘤中约 90% 为胰腺导管腺癌及其变异亚型^[1-2], 手术依然是唯一可能治愈的手段^[3]。术后病理标本中肿瘤浸润范围、切缘状态等各项病理学指标的评估是评判肿瘤预后的最关键因素, 而完成这一精准评估的前提是对胰腺切除标本精准全面的取材和规范化病理报告。这不仅有助于临床医师对患者进行个体化、高质量的管理, 而且可以为临床研究提供更准确和详尽的病理数据。此外, 新辅助治疗引起的肿瘤退缩已被证明可提高肿瘤可切除性和(或)切缘阴性(R_0 切除)的概率, 进而提高患者无病生存率和总生存率^[4-10], 因此, 经过新辅助治疗的胰腺癌手术标本也越来越多。然而到目前为止, 国内外尚无为新辅助治疗后胰腺癌手术切除标本制定规范化的病理学检查和报告指南, 这也导致不同医疗机构之间对肿瘤退缩和切缘状态等病理学指标评估的差异^[11-12]。

为规范我国胰腺癌病理诊断, 由中国医师协会胰腺病专业委员会、国家消化病临床医学研究中心(上海)和《中华胰腺病杂志》牵头, 组织病理学、外科学、内科学、影像医学、循证医学等领域专家, 遵循《世界卫生组织指南制定手册》^[13]、美国医学科学院提出的临床实践指南的定义^[14]、中华医学会发布的《中国制订/修订临床诊疗指南的指导原则(2022 版)》^[15]以及世界卫生组织颁布的《国际疾病分类(第 11 版)》^[16], 参考指南研究与评价工具(appraisal of guidelines for research and evaluation, AGREE II)^[17]和国际实践指南报告标准(reporting items for practice guidelines in healthcare, RIGHT)^[18], 依据已发表的文献, 广泛征询专家意见, 采用改良 Delphi 法, 通过多轮投票与集体讨论的方式, 针对胰腺癌病理取材及诊断报告中的关键问题, 共形成 11 条推荐意见。由海军军医大学第一附属医院病理科成立指南起草小组对指南进行起草和修订, 最终制定了《中国胰腺癌病理取材及诊断报告规范循证学指南》。本指南已在国际实践指南注册平台(<http://www.guidelinesregistry.cn/>)注册(PREPARE-2023CN491)。本指南基于推荐意见分级的评估、制定及评价(grading of recommendation assessment, development and evaluation, GRADE)方法^[19], 将证据质量分为 4 级, 依次为高(A)、中(B)、低(C)、非常低(D)。推荐意见分为强、弱 2 个级别。

一、胰腺手术标本的切缘如何定义?

推荐意见 1: 胰十二指肠切除标本手术切缘包括消化道远近端切缘、胰腺切缘、肠系膜上动脉(superior mesenteric artery, SMA)切缘、胆管或肝

管切缘; 环周表面包括前表面和后表面及肠系膜上静脉(superior mesenteric vein, SMV)沟表面。胰体尾切除标本手术切缘包括胰腺切缘及环周的前、后表面。SMV 血管沟及前、后表面不认为是真正意义上的手术切缘^[20-23]。对所有切缘及表面的取材推荐整个面广泛全取材。

证据质量: 中

推荐强度: 强

规范胰十二指肠及胰体尾手术切除标本的切缘命名, 明确真正意义上手术切缘及相应环周表面的定义, 并规范切缘取材方式, 对于更好地评判切缘状态并提供预后信息至关重要^[20-21]。手术切缘的评估主要基于大体观察和显微镜下验证。考虑肿瘤累及切缘可能在大体上难以明确, 因此切缘面广泛取材对于判定切缘状态非常重要。

1. 胰十二指肠切除标本: 胰十二指肠切除标本切缘(图 1)中, SMA(腹膜后或钩突)切缘是最重要的切缘, 是与 SMA 直接接触的 3~4 cm 范围的软组织。这个区域外周神经丰富且紧邻 SMA, 肿瘤的局部复发也多见于瘤床的钩突边缘区, 因此 SMA 切缘状态对于评估预后非常重要^[23]。本指南推荐将 SMA 切缘涂色, 并垂直于涂色面对整个平面进行全取材并行组织学检查, 有助于更清晰地显示肿瘤距切缘最近的距离。

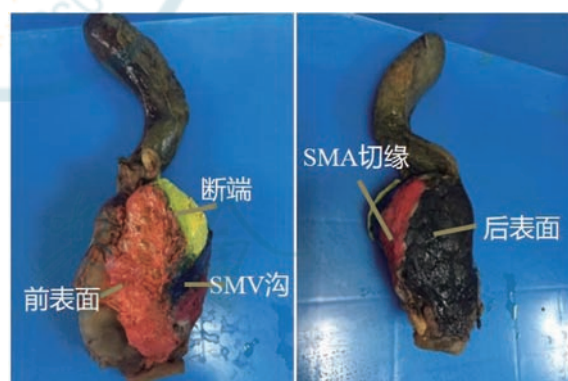


图 1 胰十二指肠切除标本切缘(墨汁涂色)。胰腺断端(黄色)、前表面(橙色)、后表面(黑色)、SMV 沟(蓝色)和 SMA 切缘(红色)。SMV 为肠系膜上静脉; SMA 为肠系膜上动脉

胰颈部(断端)切缘是胰腺断端整个面的切缘, 即沿着胰腺横断平面全取材, 断面朝下放置, 如果切片中看到癌组织, 则判定为 R_1 (0 mm)^[20-21]。但这一切缘取材及判定方式与目前的“1 mm”原则有冲突, 所以日常工作中也可以参照 SMA 切缘取材的方式。如果肿瘤肉眼上离切缘很近(≤ 1 cm), 可对整个切缘进行涂色, 然后面向涂色面作垂直切面, 且所有带涂色



切缘的组织均进行组织学检查,并以毫米级的精准度准确记录肿瘤与此切缘的距离。

胆管或肝管切缘是胆管断端的整个断面,即沿着胆管断面全取材,断面朝下放置,如果切片中看到癌组织,则判定为 $R_1(0\text{ mm})^{[20-21]}$ 。也可参照 SMA 切缘取材方式行垂直切缘取材。

消化道远近端切缘是消化道离断缘的整个断面,即沿着消化管断面全取材,断面朝下放置,这样第一张切片显示的为真正的切缘,如果切片中看到癌组织,则判定为 $R_1(0\text{ mm})^{[20-21]}$ 。也可参照 SMA 切缘取材方式行垂直切缘取材。

门静脉切缘及肿瘤侵犯门静脉管壁的深度与预后相关^[24],如果手术标本中附带部分或完整的门静脉,需特别标注。门静脉远近端切缘需要取材。取肿瘤侵犯门静脉的组织块时,应尽量包含门静脉的全层。

其他环周表面中后表面(不包括肠系膜上动脉切缘)是肠系膜上动脉切缘至胰十二指肠沟之间整个胰头后方的疏松结缔组织。后表面的取材是完整的一个面,而不是局部一个垂直面,这样有助于更清晰地明确肿瘤距切缘最近的距离。这一表面有时与 SMA 切缘在同一个组织块,因此,各个切缘、表面最好用不同颜色墨汁标记。SMV 沟是一个血管沟的光滑面或压迹(也被称为门静脉沟切缘),这是胰头后内侧的一个平滑的沟状平面,其上为 SMV。这一表面的取材也是完整的一个面,而不是局部一个垂直面,可参照 SMA 切缘取材的方式。前表面定义为从 SMV 切缘延伸至胰十二指肠沟前方的整个胰头前方的区域,这并非真正意义上手术切缘,但如果肿瘤侵犯这一表面,提示与肿瘤局部复发及患者较短的生存期相关,因此建议进行取材^[25-30]。部分患者前表面与其他手术切除的组织结构粘连,应将这些结构视为附加的环周缘,并报告与肿瘤最近的距离。

以上所有切缘和胰腺表面组织构成胰头标本的环周面。对于各个切缘和表面使用指定颜色的墨汁进行涂色有助于显微镜下识别^[20]。大体标本需要记录肿瘤与最近切缘或表面的距离,并于显微镜下明确。

2. 胰体尾切除标本:胰体尾切除标本切缘(图 2)中,如果肿瘤肉眼上离近端胰腺(断端)切缘很近($\leq 1\text{ cm}$),整个切缘可以涂色,然后面向涂色面行垂直切面,所有带涂色切缘的组织均进行组织学检查,并以毫米级的精准度去准确记录肿瘤与此切缘的距离。如果肿瘤与切缘距离较远,可采用平行切缘取材方式。



图 2 胰体尾切除标本切缘(墨汁涂色)。胰腺断端(橙色)、前表面(黑色)以及后表面(黄色)

前表面切缘反映了肿瘤与胰腺前缘软组织的关系,离肿瘤最近平面的组织需要取材以明确有无肿瘤侵犯,确切的取材数目取决于肉眼上观察的肿瘤浸润范围。

后表面切缘反映了肿瘤与胰腺后缘软组织的关系,离肿瘤最近平面的组织需要取材以明确有无肿瘤侵犯,确切的取材数目取决于肉眼上观察的肿瘤浸润范围。

胰体尾切除标本需要评估胰颈断端及胰周覆盖软组织被肿瘤累及的情况。另外脾血管及脾脏的累及情况也应记录。脾静脉及脾动脉断端切缘也应取材并进行组织学检查。需记录肿瘤与最近切缘或表面的距离,并在显微镜下明确。

二、胰腺手术标本的取材方式如何选择?

推荐意见 2:胰十二指肠切除标本推荐使用垂直于十二指肠长轴的连续横断面取材方式,这一取材方式只针对胰头癌,不推荐用于壶腹部及胆总管癌。胰体尾切除标本推荐使用沿肿瘤最大径做前后向连续切面(可垂直或平行于主胰管)。新辅助治疗后胰腺癌标本评估完全反应时推荐全取材。

证据质量:中 推荐强度:强

目前有多种关于胰十二指肠切除标本的取材方式^[22-30-34],但本指南优先推荐垂直于十二指肠长轴进行连续平行剖面^[30](图 3)。这种取材方式容易操作,无需打开胆总管及主胰管,且能在同一平面展示重要的解剖结构(如壶腹部、胆总管及主胰管),因而常被一些癌症中心作为共识推荐^[35-36]。该取材方式通常将胰十二指肠切除标本解剖成 8~10 个剖面,可详细检查肿瘤情况及其与周围关键解剖结构及切缘的关系^[31]。其最大的优势在于不仅可以从 3 个维度测量肿瘤的范围,还可以在每个剖面详细检查肿瘤与各个切缘的关系,准确判断肿瘤距切缘的距离^[37-40]。

胰腺远端切除标本中,脾动脉和脾静脉沿胰腺后上缘走行,据此可以明确前、后缘并墨汁染色,有助于标本定位。整个标本可做前后向连续切面,可平行于主胰管(4A)或垂直于主胰管(4B)。全胰切



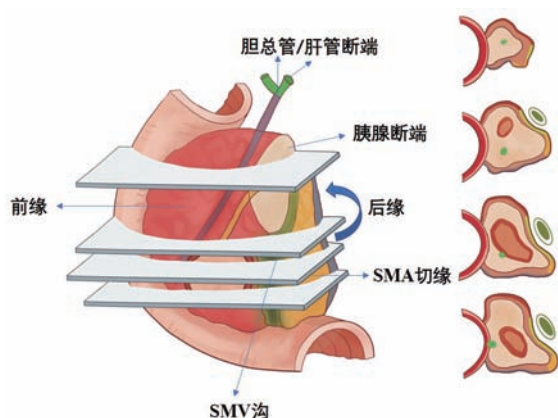


图3 胰十二指肠切除标本取材方式示意图。SMA为肠系膜上动脉;SMV为肠系膜上静脉

除标本中,胰头部可按横断面,即垂直于十二指肠长轴连续切面,而体尾部可平行于主胰管或垂直于主胰管。

标本解剖前,先取胰腺断端、胆总管断端、消化管两断端切缘。组织块需包含肿瘤累及的解剖学结构(如十二指肠、壶腹部、胆总管及胰周软组织等),同样也需包含肿瘤与各个相邻切缘的关系。但肉眼通常很难准确辨认肿瘤组织浸润范围,因此需对肿瘤及相邻的切缘广泛取材^[38,41-42]。如果条件允许,可使用大组织块包埋,有助于明确肿瘤与周围组织结

构及切缘的关系,并有助于更精准测量肿瘤的最大径^[35]。新辅助治疗后肿瘤大部分可被纤维组织所取代,大体标本很难区分肿瘤、肿瘤退缩引起的纤维化区及梗阻性胰腺炎的纤维化区^[43]。为了精准评估残存肿瘤的范围及其与切缘的关系,广泛取材是必须的。特别对于新辅助治疗后肿瘤完全反应的评判,必须整个胰腺全取材^[35,44]。此外,大体标本连续切面图像的保存(图5)可有助于后续多学科讨论中对于肿瘤来源等信息的明确。

三、胰腺癌组织学类型及分化程度如何评估?

推荐意见3:胰腺癌组织学类型推荐使用世界卫生组织(World Health Organization, WHO)第5版胰腺肿瘤分类,胰腺导管腺癌组织学分化程度推荐使用美国病理学家协会(College of American Pathologists, CAP)评估标准。当胰腺导管腺癌合并前驱病变时,需明确导管腺癌是否由前驱病变进展而来,并明确组织学亚型及分化。

证据质量:低 推荐强度:弱

本指南所述胰腺癌主要包括胰腺导管腺癌、腺泡细胞癌及前驱病变进展而来的癌,WHO第5版消化系统肿瘤分类对这几类胰腺癌的组织学类型作出了详细的阐述^[1]。

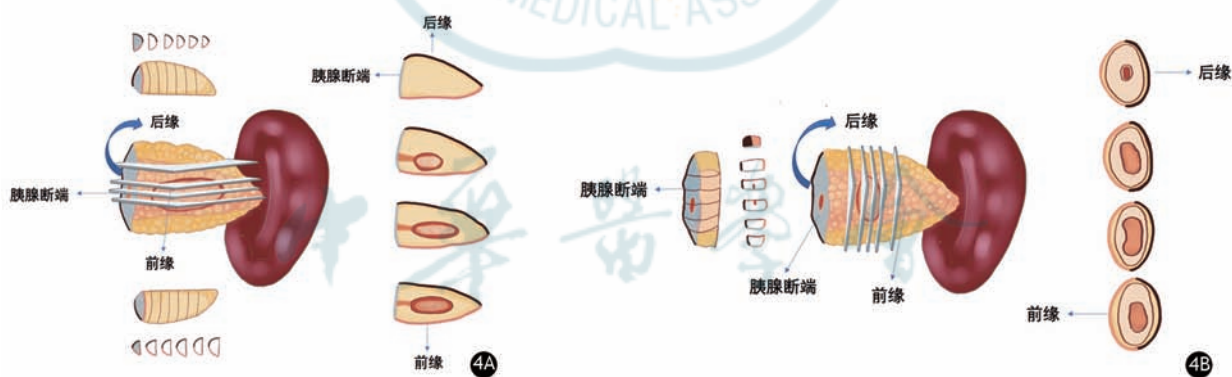


图4 胰体尾切除标本取材方式示意图。4A 平行于主胰管行连续平行剖面;4B 垂直于主胰管行连续平行剖面

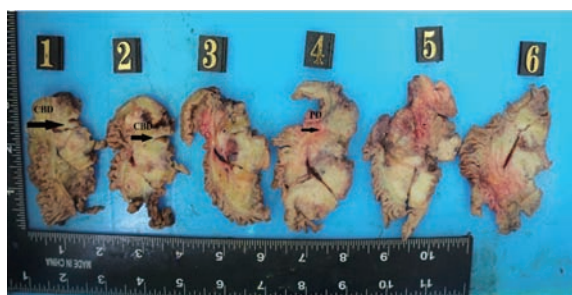


图5 胰腺大体标本连续切面拍照。CBD为胆总管;PD为胰管

胰腺导管腺癌及其特殊亚型约占胰腺恶性肿瘤90%,而腺泡细胞癌占成人胰腺恶性肿瘤不足2%。不同胰腺导管腺癌特殊亚型在生物学行为、预后和治疗策略方面均有不同,因此对其正确诊断至关重要。当胰腺导管腺癌背景中存在导管内乳头状黏液性肿瘤(intraductal papillary mucinous neoplasm, IPMN)、导管内嗜酸细胞性乳头状肿瘤(intraductal oncocytic papillary neoplasm, IOPN)、导管内管状乳头

状肿瘤 (intraductal tubulopapillary neoplasm, ITPN)、黏液性囊性肿瘤 (mucinous cystic neoplasm, MCN) 这些前驱病变时, 诊断中一个非常关键的要点就是明确浸润性癌的成分是否为导管内肿瘤 (IPMN、ITPN、IOPN) 或者 MCN 进展而来^[45-46]。MCN 进展而来的导管腺癌 5 年生存率要明显优于非 MCN 进展而来的导管腺癌^[45-46]; 无论是 IPMN 进展而来的导管腺癌或 IPMN 合并导管腺癌, 相较普通型导管腺癌均有更好的预后, 且临床也更易早期诊断。在浸润性癌亚型方面, IPMN 来源的胶样癌预后优于 IPMN 来源的导管腺癌^[47-50], IOPN 进展而来的嗜酸细胞癌是所有亚型中预后最好的, 5 年疾病特异性生存率达 100%^[48,51-52]; ITPN 伴浸润性癌患者 5 年生存率约 71%, 明显高于普通型导管腺癌^[53-56]。其他少见亚型包括透明细胞型^[57-59]、嗜酸细胞亚型^[60]、纤毛细胞型^[61]、富于空泡型^[62]、肠型^[63]、微腺型^[64]、囊性型^[65]、大导管型^[66]、囊性乳头状型^[67]、黏液表皮样癌^[68]及非黏液性糖原缺乏性囊腺癌^[69], 且随着分子病理学的进展, 近年还报道了伴 *SMARCB1/INI1* 缺失相关的胰腺癌^[70-71], 由于这些肿瘤类型少见, 研究资料有限, 其临床病理学意义有待更深入的研究。

组织学分化对于胰腺导管腺癌具有很好的预后评判意义^[72], WHO^[1]及 CAP^[73]都给出了相应的评估标准, 相互之间一致性很高, 也有相似的预测价值^[74]。CAP 相较于 WHO 更为简单, 无需评估黏液产生及核分裂, 故应用更广泛, 因此本指南对于胰腺导管腺癌的组织学分化也推荐使用 CAP 系统: 1 级, >95% 的肿瘤形成腺样结构; 2 级, 50%~95% 的肿瘤形成腺样结构; 3 级, <50% 的肿瘤形成腺样结构。

四、淋巴管或血管侵犯如何评估?

推荐意见 4: 淋巴管或血管侵犯应单独记录, 对于已命名的血管 (腹腔干、肠系膜上动脉、肝总动脉、门静脉、肠系膜上静脉、脾动脉、脾静脉) 应记录其侵犯深度 (外膜、中膜、内膜)。

证据质量: 中 推荐强度: 强

无论胰腺癌患者是否接受新辅助治疗, 淋巴管、血管侵犯均是不良预后的评价指标^[75-77]。根据美国癌症联合会 (American Joint Committee on Cancer, AJCC)/国际抗癌联盟 (The Union for international Cancer Control, UICC) TNM 分期第 8 版及美国国立综合癌症网络 (The National Comprehensive Cancer Network, NCCN) 指南, 这些病理学特征均应被单独记录。

对于肿瘤内小的淋巴管及静脉, 常规切片中很

难区分, 出现伴随的小动脉可帮助识别静脉侵犯, 特殊染色特别是弹力纤维染色或 CD₃₁、CD₃₄ 和 D2-40 免疫组织化学标记有助于鉴别诊断。同时, 免疫组织化学染色或特殊染色也可识别那些误认为胰腺上皮内瘤变 (pancreatic intraepithelial neoplasia, PanIN) 的脉管侵犯病变。大的已命名的血管侵犯对于胰腺癌患者术后的复发和 5 年生存期的预测具有重要价值, 尤其是影响 TNM 分期的腹腔干、肠系膜上动脉及肝总动脉要明确标明有无侵犯并标注侵犯深度^[23]; 对于还未纳入 T 分期的门静脉、SMV、脾动脉、脾静脉侵犯, 也建议记录侵犯深度, 因为其对肿瘤转移复发及预后的预测仍有一定价值^[24,78-85]。

五、外周神经侵犯如何评估?

推荐意见 5: 推荐报告外周神经侵犯情况, 并建议区分胰内神经侵犯和胰周神经侵犯。

证据质量: 低 推荐强度: 弱

外周神经侵犯是指肿瘤细胞至少侵犯了神经纤维 3 层结构中的任意一层或肿瘤细胞包绕神经纤维 1/3 以上^[86]。外周神经侵犯是胰腺癌重要的组织学特征, 同时也是胰腺癌术后复发及预后不良的重要因素^[77,87-91]。胰腺外周神经分布丰富, 分为胰腺被膜以内的胰内神经和被膜以外的胰周神经。胰内神经侵犯是影响术后胰腺癌患者预后及复发的重要因素^[88,92-93], 且胰内神经侵犯与胰周神经侵犯具有明显的相关性^[94], 胰内神经侵犯导致癌细胞沿神经向胰周神经及软组织侵犯^[95-96], 胰周神经侵犯的患者更易复发且预后更差^[97-99]。外周神经侵犯在经过新辅助治疗的患者中也是一个不良预后指标^[100-101]。因此本指南推荐将胰内神经侵犯及胰周神经侵犯分别记录。

六、胰腺癌 R₁ 切缘如何评估?

推荐意见 6: R₁ 定义为显微镜下肿瘤距切缘距离 ≤1 mm。

证据质量: 非常低 推荐强度: 弱

虽然目前国际各大指南对 R 状态的评估主要基于对 SMA 切缘、胰腺切缘、胆管切缘、消化道远近端切缘这些真正手术切缘的评估^[20-21,73,102], 但均推荐对切除标本的所有表面进行评估, 包括胰腺前表面、后缘和 SMV 血管沟表面^[21,102], 因为肿瘤累及这些表面增加了局部肿瘤复发的风险, 并与预后相关^[26,40,103-109]。较多基于全面详细病理检查方案的研究发现, 通过评价所有表面提示较高 R₁ 率 (>70%), 并与生存相关^[27,34,38-39,110-114]。因此, 本指南推荐对于所有切缘及表面均应进行仔细、全面、详尽地取材, 并在显微镜下

明确肿瘤距切缘或表面最近的距离。

胰腺癌 R_0 与 R_1 的界定仍然存在争议。该定义最初来源于直肠癌评估方案,发现肿瘤距切缘 ≤ 1 mm 可预测局部复发并提示更差的生存期^[115-119]。考虑胰腺导管腺癌更为分散的生长模式,有研究提示更大的距离(如 1.5 mm)预测预后的价值更高^[119-121],但并未在诊断实践中应用。另外从解剖学上来说胰腺的前表面是腹膜覆盖组织,因此前表面的受累应定义为表面的突破,也就是 0 mm^[21,122]。

少数情况下,切缘 1 mm 以内的胰腺实质及间质纤维、脂肪组织内未见明确的肿瘤,而在这一范围内的淋巴结、淋巴管、血管及外周神经周围查见肿瘤,此时是否将切缘定义为 R_1 ,目前尚无循证医学证据去阐释。UICC 的 TNM 分期中提出,如果在切缘 1 mm 范围内的脉管腔内查见肿瘤细胞,此时归为 R_0 ,而当肿瘤贴附于脉管壁或侵犯脉管壁时,此时归为 R_1 ^[110]。对切缘 1 mm 以内淋巴结及外周神经受累如何定义,目前缺乏证据,建议在报告中记录说明。

另外, $R_1 \leq 1$ mm 的定义仅适用于胰腺导管腺癌。没有证据表明这一定义也适用于腺泡细胞癌,因为腺泡细胞癌与胰腺导管腺癌不同,通常不呈分散的生长模式,因此建议记录肿瘤距切缘最近的距离。

接受新辅助治疗患者 R_1 切缘距离的界定目前仍缺乏共识^[44]。首先,在新辅助治疗后,点取样可能会导致切缘评估出现假阴性结果。其次,判定 R_1 的肿瘤细胞与切缘之间距离也存在争议。有研究提出肿瘤距切缘最近距离 > 2 mm 提示更好的预后^[123-124]。因此,对于新辅助治疗后的手术切除标本,本指南推荐采用标本全取材和大组织病理切片对切缘进行全面的评估,并详细记录残留病灶距离切缘的最近距离。

七、T 分期如何评估?

推荐意见 7: T 分期参照 AJCC/UICC 第 8 版分期系统。IPMN、IOPN、ITPN、MCN 伴浸润性癌, $T_1 \sim T_3$ 分期评估标准以浸润灶的最大径为准,而非整个病变的大小。新辅助治疗后胰腺手术切除标本推荐采用大切片取材方式并记录测量肿瘤大小方法。

证据质量:低

推荐强度:弱

T 分期参照 AJCC/UICC 第 8 版分期系统^[23],腺泡细胞癌分期也参照胰腺导管腺癌。“m”代表存在多个原发肿瘤,“r”代表复发性肿瘤,“y”代表经过新辅助治疗。

肿瘤大小是肿瘤非常重要的特征,反映了原发肿瘤负荷,因此是患者预后最强的预测因子^[125]。由于胰腺癌的生长方式高度分散、胰腺实质萎缩和纤

维化,肉眼很难明确真正的肿瘤边界^[119],目前大多数指南推荐使用肉眼测量肿瘤大小并在显微镜下确认^[11,35,73,126-127]。T 分期以最大肿瘤直径作为评估标准。相较于肿瘤侵犯胰周的范围,肿瘤的大小可更好地进行预后分层^[108,128-131]。根据肿瘤最大径分为 T_1 (最大径 ≤ 2 cm)、 T_2 (最大径 > 2 cm 且 ≤ 4 cm)、 T_3 (最大径 > 4 cm), T_4 评判标准为肿瘤侵犯肠系膜上动脉或腹腔干或肝总动脉,不论肿瘤大小^[23]。 T_1 还可进一步分为 T_{1a} (最大径 ≤ 0.5 cm)、 T_{1b} (最大径 > 0.5 cm 且 < 1 cm)、 T_{1c} (最大径 ≥ 1 cm 且 ≤ 2 cm)。 T_{is} 包括 PanIN3、IPMN 伴上皮高级别异型增生、ITPN 伴上皮高级别异型增生、IOPN 伴上皮高级别异型增生及 MCN 伴上皮高级别异型增生。

T 分期需要注意的是,如果存在 1 个以上肿瘤,应详细记录肿瘤的个数、位置及每个肿瘤的大小,但这只适用于大体可识别的多个肿瘤,不适用于肉眼为单个、而镜下呈多灶性分布的肿瘤^[77]。对于胰腺导管腺癌的前驱病变伴浸润性癌,包括 IPMN、IOPN、ITPN、MCN 伴浸润性癌, $T_1 \sim T_3$ 分期评估标准以浸润灶的最大径为准,而非整个病变的大小^[46,132]。如果浸润灶是多灶性的,T 分期评估标准是以最大浸润灶直径还是以多个浸润灶的总和为准,目前尚存争议。因此在日常工作中尽量评估详尽,对于浸润灶的多少、每个浸润灶的最大径及整体浸润灶的最大径均应进行评估,为后续临床研究留下详尽病理资料,以期达成一个更精准及规范的 T 分期标准。

对于新辅助治疗后胰腺癌切除标本,目前的测量方法存在一定的局限性。新辅助治疗后肿瘤细胞退缩并被纤维组织取代,特别在肿瘤细胞绝大部分退缩的情况下,肿瘤周围胰腺组织萎缩纤维化严重,增加了肉眼准确辨别肿瘤边界的难度。目前的 T 分期标准很难准确反映患者的预后情况^[133],因此需要更加科学的手段去评估新辅助治疗后肿瘤的实际大小。有研究者利用显微镜下确认肿瘤边界,对比大体图片描绘肿瘤边界并测量肿瘤大小,从而进行 T 分期的评估^[134],虽比肉眼直接测量更加准确,但操作过程繁杂,仍存在较大的误差。目前主要推荐以下两种方式进行肿瘤大小评估:(1)显微镜下测量整个区域内残留存活肿瘤细胞的最大线性直径,包括肿瘤细胞之间非癌组织,如间质和(或)胰腺实质或其他组织结构。(2)显微镜下测量每个肿瘤灶的最大径,并计算其总和。无论哪种测量方式,精准测量的基础都依赖于精准取材,且能否反应肿瘤的最大范围。因此本指南推荐对于新辅助治疗后胰腺手术切除标



本采用大切片取材方式以最大限度展示肿瘤全貌及与周围组织关系,并在病理报告中记录显微镜下大小测量方法。

肿瘤侵犯周围组织包括侵犯胰周软组织、腹膜(包括结肠系膜、大小网膜)、胰外胆管系统,胰头部肿瘤还可侵犯十二指肠(包括壶腹部),胰体尾肿瘤还可侵犯胃、横结肠、脾脏及左侧肾上腺。肿瘤侵犯周围组织虽不影响肿瘤分期,但也应在病理报告中记录。

八、N 分期如何评估?

推荐意见 8: 参照 AJCC/UICC 第 8 版分期系统,肿瘤直接侵犯与转移均计入阳性淋巴结个数。淋巴结检出尽可能充分,Whipple 手术标本淋巴结检出量至少应达 12 枚。

证据质量:中 推荐强度:强

淋巴结状况是预测胰腺导管腺癌生存最有效的指标之一,基于预后数据,根据阳性淋巴结个数分为 N_0 (无区域淋巴结阳性)、 N_1 (1~3 枚区域淋巴结阳性)和 N_2 (4 枚或更多的区域淋巴结阳性)。切除标本的所有淋巴结均应做组织学检查,单独送检的区域淋巴结应单独报告。

阳性淋巴结的检出数量已被证明与患者生存期相关,而淋巴结取样不充分可导致 N 分期不足^[135-139]。国际癌症报告合作组织(International Collaboration on Cancer Reporting, ICCR)、CAP 及 AJCC/UICC 第 8 版提出 Whipple 手术标本淋巴结检出量至少应达到 12 枚。基于多项研究数据,英国皇家病理学家学会认为 Whipple 手术标本淋巴结检出量至少应达到 15 枚^[137-138,140-141]。新辅助治疗后的患者,检出淋巴结数目常下降,可低于 15 枚^[142]。远端胰腺切除标本应检出最小淋巴结数量尚未形成共识,是否进行扩大淋巴结清扫目前仍有争议。

肿瘤直接侵犯淋巴结可在高达 20% 的胰腺癌切除标本中被发现。有研究认为,直接侵犯并不代表真正的淋巴结转移(即通过淋巴管扩散),在预后上等同于 $pN0$ ^[143-145]。也有研究表明,直接侵犯与“真正的” $pN1$ 切除的患者预后相当^[146-147]。在 UICC/AJCC 第 8 版分期系统中,肿瘤直接侵犯的淋巴结也被纳入到阳性淋巴结^[102,148]。另有研究认为阳性淋巴结比例(阳性淋巴结数量与检出的淋巴结总数之比)也是有效的预后指标,阳性淋巴结比例 >20% 与患者较低的生存率显著相关^[149-156]。

在 AJCC/UICC TNM 和日本胰腺学会(Japan Pancreas Society, JPS)系统中,肝总动脉周围的淋巴

结被认为是区域淋巴结^[102],而主动脉旁淋巴结并不认为是区域淋巴结,主动脉旁淋巴结阳性患者的生存率明显低于阴性患者,转移到主动脉旁淋巴结被认为是远处转移(即 $pM1$)^[157-159],但这尚未被采用到临床实践中。

九、M 分期如何评估?

推荐意见 9: 无远处转移为 M_0 ,有远处转移为 M_1 ,腹膜种植或腹水阳性均诊断为 M_1 。

证据质量:中 推荐强度:强

M 分期最值得注意的是腹膜种植或腹水阳性,均被诊断为 M_1 ^[23,160]。

十、新辅助治疗后肿瘤退缩分级(tumor regression grading, TRG)如何评估?

推荐意见 10: 推荐使用 CAP 和 MD 安德森癌症中心(MD Anderson Cancer Center, MDACC)评估系统对新辅助治疗后 TRG 进行评估。

证据质量:低 推荐强度:弱

胰腺癌新辅助治疗在改善胰腺癌患者预后中的作用越来越受到重视,在对新辅助治疗后的胰腺癌手术切除标本进行病理评估时,通常会评估新辅助治疗后 TRG,以评估肿瘤对接受的新辅助治疗的敏感性,从而指导后续辅助治疗并预测患者预后。目前有多种 TRG 评估系统用于胰腺癌,包括最早于 1992 年提出的 Evans 系统^[161]、CAP 评估系统^[162](基于 Ryan 等^[163]提出的最初用于评估直肠癌新辅助治疗反应的分级系统,又称为改良的 Ryan 方案)、基于 Chatterjee 等^[164-165]提出的 MDACC 评估系统和 JPS 评估系统^[30](表 1)。虽然目前未形成最佳的评估系统共识,但可以肯定的是完全和接近完全反应的患者预后较好^[164]。CAP 评估系统被 NCCN 指南推荐;Evans 和 JPS 评估系统专用于胰腺导管腺癌,在日本常用。目前中国胰腺癌诊治指南^[166]推荐使用 CAP 评估系统。MDACC 评估系统与 CAP 评估系统类似,但采用三分类而不是四分类。以上 4 种 TRG 评估系统均为半定量,除肿瘤完全退缩评估标准相同外,其余评估肿瘤消退程度的界值和标准都不一样,因此很难将它们进行比较。

肿瘤完全退缩、接近完全退缩、部分退缩及无退缩是新辅助治疗的重要疗效指标,但其组织学评估是否可靠一直是值得关注的问题。困难主要在于如何建立可靠的完全退缩的诊断标准及可重复的不同退缩程度的评估标准,因此广泛全面的组织取样至关重要。但是,大多数研究缺乏有关取材范围的信息;少数已发表的数据表明研究内部和研究之间存在



表 1 国际常用的 4 种胰腺癌肿瘤退缩分级评估系统

肿瘤退缩分级 评估系统	肿瘤退缩 分级	定义
Evans	Grade I	无或<10%肿瘤细胞损毁
	Grade II a	10%~50%肿瘤细胞损毁
	Grade II b	50%~90%肿瘤细胞损毁
	Grade III	极少(<10%)存活肿瘤细胞
	Grade IV	无存活肿瘤细胞
CAP	Score 0	无存活肿瘤细胞
	Score 1	单个或小灶性肿瘤细胞残留
	Score 2	残留肿瘤灶伴间质纤维化
	Score 3	少数或无肿瘤退缩;大量肿瘤细胞残留
JPS	Grade 1a	估计肿瘤残留率≥90%
	Grade 1b	估计肿瘤残留率≥50%,<90%
	Grade 2	估计肿瘤残留率≥10%,<50%
	Grade 3	估计肿瘤残留率<10%
	Grade 4	无存活肿瘤细胞
MDACC	Score 0	无存活肿瘤细胞
	Score 1	<5%存活肿瘤细胞
	Score 2	≥5%存活肿瘤细胞

注:CAP为美国病理学家协会;JPS为日本胰腺协会;MDACC 为 MD 安德森癌症中心

重大差异。一项基于胰腺和周围组织完全包埋的研究表明仅 2.5% 的患者出现肿瘤完全退缩^[167],而其他没有进行广泛取样的研究结果提示,10%~33% 患者出现肿瘤完全退缩^[168-177]。因此,肿瘤退缩的准确评估需要对病变组织广泛取样,特别对于肿瘤完全退缩的评估,需要对整个肿瘤床和任何相邻的肉眼可见异常的组织全取材并进行组织学检查。

十一、胰腺癌背景病变需要评估吗?

推荐意见 11: 需要评估胰腺癌背景病变如 PanIN、IPMN、ITPN、IOPN、MCN 及慢性胰腺炎。

证据质量:低 推荐强度:弱

除了关注肿瘤本身,肿瘤背景病变的认识也越来越受到重视。PanIN 是胰腺导管腺癌最常见的前驱病变^[178],有研究者发现,不伴有 PanIN 的胰腺癌患者术后存活率低^[179]。若同时合并 IPMN、ITPN、IOPN 或 MCN,也需在病理报告中标注,并明确上皮异型的程度及与胰腺癌的关系^[132,180]。NCCN 指南中也提出需要在病理诊断报告中标注出背景病变例如有无 PanIN 或慢性胰腺炎。

目前尚无针对胰腺癌病理取材及诊断报告规范指南,本工作小组结合国内外现有证据,提出 5 个强烈推荐意见和 6 个弱推荐意见,有利于对胰腺癌进行精准分期、生存预测和指导术后治疗,有助于提升病理

报告完整性,加强病理医师与临床医师之间的沟通,同时也提高了各医院病理报告的同质化水平,因此具有较强的推广价值。

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